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Adrenocortical steroids, bone mineral content and endometrial condition in post-menopausal women

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In 23 post-menopausal women, serum levels of cortisol, unconjugated dehydroepiandrosterone (DHA), dehydroepiandrosterone sulphate (DHAS), testosterone, unconjugated and total oestrone and prolactin were measured before and during an ACTH test. Significant positive correlations were found between basal levels of DHA and DHAS; DHA and unconjugated oestrone; DHA and total oestrone; testosterone and total oestrone and between unconjugated and total oestrone. ACTH significantly raised the levels of the steroids but not of prolactin. Significant positive correlations were found between basal levels and ACTH induced increments in DHA; between basal DHAS and increments in DHA and between increments in DHA and DHAS. A significant negative correlation was found between basal levels and increments in cortisol. No significant correlations were found between other combinations of hormone basal levels and/or increments. Significant positive correlations were found between basal levels of DHAS and the DHA response to ACTH respectively, and trabecular bone mineral content of the distal forearm. A significant correlation was also found between bone mineral content and pre-cancerous/cancerous state of the uterine epithelium.

The results are a further support to the concept of a link between adrenal androgens and bone mineral density, and do also indicate a relation to endometrial pathology. The lack of correlation between cortisol and other steroids indicate different regulatory mechanisms. Prolactin does not seem to be involved in the regulation of the adrenal androgen synthesis.

(Key words: Adrenal cortex, Androgens, Oestrogens, ACTH, Bone mineral content, Endometrial condition)

Introduction

Oestrogen deficiency is generally considered as one of the two main risk factors in the genesis of post-menopausal osteoporosis ([1–4], for further reference see [5]). An association has been demonstrated between bone loss/osteoporosis and low peripheral serum levels of 17β-oestradiol [1], oestrone and 4-androstene-3,17-dione [2,3]. On the other hand endometrial carcinoma represents a condition associated with high peripheral levels of oestrogens [6–9] but also of dehydroepiandrosterone sulphate [9], 4-androstene-3,17-dione [10] and testosterone [11]. Adrenal hyperplasia and the polycystic ovary syndrome are two clinically hyperandrogenic conditions associated with endometrial carcinoma [12]. Besides the common hypothesis of an association between unopposed oestrogen influence and endometrial cancer, the role of androgens in this respect has also been discussed [9].

In the past these different pathologic conditions have been studied separately in the oophorectomized or post-menopausal women. In the present paper we have attempted to consider these disease entities in a broader context. Our aim has been to create a more integrated picture of the relationship between serum steroid concentrations and expressions of metabolic (osteoporosis) or endometrial pathology. As a part of this study we have chosen not only to analyse basal levels of a number of steroids, the only method used in this respect up to date, but also to introduce a more dynamic approach by means of measuring several steroid hormones before and during adrenal stimulation. Furthermore, we have considered it to be of interest to study the interrelationship between these different pathologic manifestations, associated with high or low peripheral steroid levels, respectively. In a preliminary study [13] we have been able to establish an inverse relationship between bone mineral density and the existence of endometrial pathology. This may indicate a future possibility of evolving criteria for predicting the proneness of individual women to develop metabolic disease with its serious sequelae or endometrial carcinoma at the other extreme of the scale. The desirability of such a diagnostic instrument is obvious in view of the contrasting therapeutic modalities in terms of oestrogen therapy applied in these different conditions.

The main source of steroids in the post-menopausal woman is the adrenal cortex, which is reported to contribute to more than 95% of the dehydroepiandrosterone and its sulphate, 70–90% of the 4-androstene-3,17-dione and 50% of the testosterone in the peripheral circulation. The ovarian stroma secretes considerable amounts of the testosterone (50%) even after the menopause, but the ovarian production of other androgens is reported to be very limited in the post-menopausal woman [14–16]. Circulating levels of oestrone and 17β-oestradiol have an almost exclusive (indirect) adrenal origin in the post-menopausal woman [15].

Increased insight in the regulation of the adrenocortical steroid biosynthesis may contribute to our understanding of the aetiology of osteoporosis and endometrial carcinoma respectively. Besides ACTH, other pituitary hormones have been ascribed a role of regulators of the adrenocortical androgen biosynthesis, one of them being prolactin [17–19]. The present communication describes the investigation of basal levels and response to ACTH stimulation of serum oestrogens, androgens, cortisol and prolactin in post-menopausal women with varying bone mineral content and varying condition of the uterine epithelium.

Material and methods

Clinical material

The clinical material comprised 23 post-menopausal women (up to 10 yr after the menopause), aged 49–62 yr (mean age 55.2 yr) and with post-menopausal levels of serum FSH. They had no clinical or laboratory signs of hepatic, biliary or renal malfunction and were not taking any medications known to interfere with the hormone values. Fifteen had a normal, atrophic endometrium, five endometrial hyperplasia/atypia and three histologically confirmed endometrial carcinoma. 0.25 mg ACTH (Synacthen®, CIBA) was given intravenously and venous blood samples were collected 15 minutes and immediately before and 90 and 120 min after the ACTH injection. The tests were performed at forenoon between 8.30 and 12.00 h. Serum samples were coded and stored at −20° until analysed. In order to minimize the influence of inter-assay variation all samples from one patient were analysed in the same assay.

Hormone analysis

Serum levels of prolactin were measured by radioimmunoassay using a commercial kit obtained from Kabi AB, Stockholm, Sweden. Bound and free [^{125}I] prolactin was separated by precipitation by polyethylene glycol. The values were expressed as μg per litre of human prolactin NIH V.L.S. 1.

Serum cortisol was determined by radioimmunoassay without prior extraction using a commercial kit from Diagnostic Products Corp, Los Angeles, Calif., USA. The method uses an [^{125}I] cortisol tracer and free and bound radioactivity is separated by a double antibody — polyethylene glycol precipitation technique. Serum testosterone was determined radioimmunologically after ether extraction using a commercial kit from Diagnostic Products Corp. Bound and free [^{125}I] testosterone was separated as described for cortisol. The antibody used cross reacts to 22% with 5α-dihydrotestosterone and to 2.1–3.8% with five other C_{19} steroids of which only androsterone is of quantitative importance. The cross reacting steroids will acount for 10–15% of the testosterone values obtained (calculated from [20]).

Serum dehydroepiandrosterone (DHA) was determined after ether extraction by radioimmunoassay using anti-dehydroepiandrosterone-17-(carboxymethyl oxime) bovine serum albumin (Hypolab SA, Coinsins, Switzerland). This antibody cross reacts to 7.3% with 5-androstene-3β,17β-diol and to 4.1% with 5-pregnenolone. These steroids together will thus account for 3–4% of the DHA values obtained (calculated from [16,21–23]).

Serum dehydroepiandrosterone sulphate (DHAS) was measured by a modification (9) of the procedure of Metcalf [24]. The method includes hydrolysis of DHAS by autoclaving diluted serum at pH 4.5 at 120°, extraction with diethyl ether and radioimmunoassay as described for unconjugated DHA. Unconjugated DHA will account for approximately 1% and the sulphates of 5-androstene-3β, 17β-diol and 5-pregnenolone for less than 1% of the values obtained (calculated from [16,21,25,26]).

Serum levels of unconjugated oestrone were determined after ether extraction by radioimmunoassay [27] using anti-oestrone-6-thyroglobulin (Miles-Yeda Ltd., Rehovot, Israel). The antibody cross reacts to 0.4% with 17β-oestradiol and to less than 0.1% with other serum steroids of quantitative importance.

Total oestrone was determined according to Carlström and Sköldefors [28]. The method includes hydrolysis of the conjugated oestrone by *Helix pomatia* enzyme, ether extraction and radioimmunoassay as described for unconjugated oestrone. The method determines the sum of unconjugated oestrone + conjugated oestrone (oestrone sulphate) hydrolyzed by *Helix pomatia* enzyme.

Within and between assay variations were as follows: for prolactin 7.2 and 14.6%; for cortisol 4.5 and 7.0%; for testosterone 5.8 and 10.1%; for DHA 5.4 and 7.1%; for DHAS 8.1 and 12.1%; for unconjugated oestrone 7.0 and 9.8%; and for total oestrone 7.0 and 8.9%, respectively.

Bone densitometry

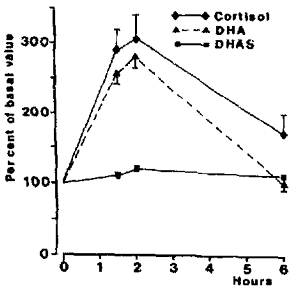
The mineral content in distal radius and ulna of the non-dominant forearm was estimated by photon absorptiometry according to Cameron and Sorensen [29] using a Studsvik Bonescanner Mod 7102 (AB Atomenergi, Studsvik, Nyköping, Sweden). This dichromatic bone densitometer measures the relative absorption in the forearm of gamma irradiation from ^{125}I and ^{241}Am, respectively [30]. The values are expressed as grams of hydroxyl apatite per centimeter.

Statistical methods

Both parametric and nonparametric tests were used according to type of distribution as specified in the results section. The significance level is set to $P < 0.05$.

Results

The first 8 ACTH tests performed included an additional sample at 6 h after ACTH injection. The change in serum steroids following ACTH injection, expressed



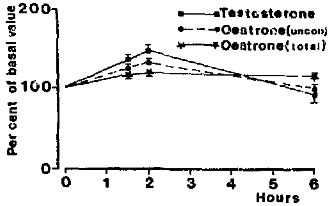


Fig. 1. Relative serum levels of cortisol, dehydroepiandrosterone (DHA) and dehydroepiandrosterone sulphate (DHAS) following an i.v. injection of 0.25 mg ACTH in post-menopausal women. The values are given as percentage of the pre-treatment value in means and SEM.

Fig. 2. Relative serum levels of testosterone, unconjugated oestrone and total oestrone as described in Fig. 1.

as percentage of the mean of the 2 pretreatment values, are given in Figs. 1 and 2. As the highest response in all steroid levels during the investigational period of 6 h was noted after 120 min, these values were used in the calculations. The mean of the values 15 min and immediately before ACTH injection is always referred to as the basal value.

Median basal serum levels and maximum levels after ACTH injection are given in Table I. Using Spearman's rank correlation test, significant correlations were found between basal values of the following hormones: testosterone and unconjugated oestrone ($r = 0.69$, $P < 0.001$); testosterone and total oestrone ($r = 0.57$, $P < 0.01$); DHA and DHAS ($r = 0.49$, $P < 0.05$); DHA and unconjugated oestrone ($r = 0.64$, $P < 0.01$); DHA and total oestrone ($r = 0.49$, $P < 0.05$); DHAS and total oestrone ($r = 0.44$, $P < 0.05$) and unconjugated and total oestrone ($r = 0.74$, $P < 0.001$). No significant correlations were found between other combinations of basal values.

ACTH administration resulted in highly significant ($P < 0.001$, t-test) increases in cortisol, DHA, DHAS, testosterone, and unconjugated oestrone 120 min after injection. The values for total oestrone were statistically significantly ($p < 0.001$) increased after 120 min; however, the change measured did not exceed the least detectable dose for the method used (0.26 and 0.33 nM, respectively). Prolactin levels were not affected by ACTH administration.

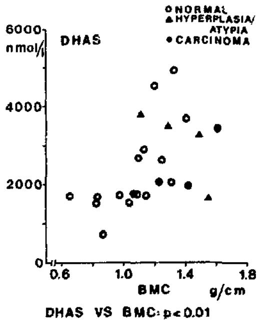
A significant negative correlation ($r = -0.42$, $P < 0.05$), Spearmans rank correlation test) was found between basal values and ACTH induced increments in cortisol, and significant positive correlations were found between basal levels and ACTH-induced increments in DHA ($r = 0.58$, $P < 0.01$), between basal DHAS and increments in DHA ($r = 0.62$, $P < 0.01$) and between the ACTH-induced increments in DHA and DHAS ($r = 0.58$, $P < 0.01$). No significant correlations were found between any other combination of basal values and increments or between increments in different hormones. It should be noted that there were no correlations between basal levels or increments in cortisol and any other steroid. This was also the case for prolactin.

When the hormone values were correlated to bone mineral content, significant

TABLE I

PRE-TREATMENT LEVELS AND MAXIMUM LEVELS AFTER ACTH INJECTION OF SERUM CORTISOL, ANDROGENS, OESTROGENS AND PROLACTIN IN POST-MENOPAUSAL WOMEN. THE VALUES ARE GIVEN AS MEDIAN AND RANGE; $n = 23$.

	Pre-treatment value	Value after ACTH injection
Cortisol (nM)	391 (178 – 800)	1063 (880 – 1410)
DHA (nM)	15.5 (7.9 – 51.8)	42.1 (22.2 – 74.0)
DHAS (nM)	2035 (758 – 5333)	2412 (1273 – 7696)
Testosterone (nM)	1.20 (0.40 – 1.95)	1.80 (0.80 – 2.60)
Unconjugated oestrone (pM)	148 (98 – 299)	188 (115 – 329)
Total oestrone (nM)	0.91 (0.43 – 3.28)	1.17 (0.53 – 3.55)
Prolactin (μg/l)	4.7 (1.8 – 7.0)	4.4 (2.8 – 6.8)



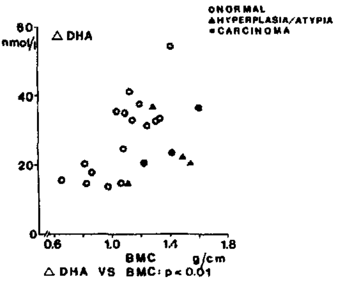


Fig. 3. Basal serum dehydroepiandrosterone sulphate (DHAS) and bone mineral content (BMC) in post-menopausal women.

Fig. 4. Increment in serum dehydroepiandrosterone 120 min after i.v. injection of 0.25 mg ACTH (Δ DHA) and bone mineral content (BMC) in post-menopausal women.

positive correlations were found between bone mineral content and basal levels of DHAS ($r = 0.61$, $P < 0.01$) and increment in DHA ($r = 0.53$, $P < 0.01$), respectively (Figs. 3 and 4; Spearman's rank correlation test). No correlation was found between bone mineral content and initial levels of cortisol, DHA, testosterone, unconjugated oestrone, or total oestrone or in the ACTH-induced increments in cortisol, DHAS, testosterone, unconjugated oestrone, or total oestrone.

An interesting observation in the present study was the finding of a significant correlation ($P < 0.05$; Kruskal–Wallis test) between bone mineral content and the condition of the uterine epithelium, e.g. normal, hyperplasia/atypia or carcinoma (Figs. 3 and 4).

There were no correlations between the women's age, weight, length or Brocas index on the one hand and basal values or increments serum hormones, bone mineral content or endometrial condition on the other.

Discussion

Our results agree with the fact that the adrenal cortex is the main source of sex steroids in the post-menopausal woman. ACTH raised the levels of all steroids studied, although the increase in total oestrone was below the sensitivity limit of the method used. Similar results following injection of ACTH to healthy post-menopausal women have previously been described by Vermeulen and co-workers [15,18]; however, in contrast to our results they found no ACTH-induced increase in DHAS in normal subjects [18].

Androgens and oestrogens increased in the same fashion as did cortisol, but with quantitative differences. Despite that, there was no correlation at all between basal levels or ACTH-induced increment in cortisol and any of the other steroids. While cortisol is mainly produced in the zona fasciculata and its synthesis is stimulated by ACTH, adrenal androgen biosynthesis occurs mainly in the zona reticularis and appears to be mediated by a dual system including ACTH and a hypothetical adrenal androgen stimulating hormone (AASH) [19]. The lack of correlation between cortisol and other steroids may reflect this different mode of regulation.

There are several reports about an association between hyperprolactenaemia and elevated values of urinary 17-oxo-steroids or serum DHA and DHAS [17–19]. Together with the finding of specific prolactin binding sites in the rat adrenal, this has given rise to the suggestion that prolactin may be the unidentified AASH. However, in the present study no correlations were found between prolactin levels and basal values or increments in any of the steroids. This favours the view expressed by Grumbach et al. [19] that prolactin is not involved in the regulation of the adrenal steroid synthesis.

The significant correlation between basal levels and ACTH-induced increments of DHA/DHAS may be due to either individual differences in the synthetic capacity of the zona reticularis and/or in its sensitivity to stimuli. The increments in DHAS were significantly correlated to the increments in DHA, and this probably reflects peripheral sulphurylation of the increased DHA levels. It has been demonstrated by Vaitukuitis and co-workers [31] that DHAS in adrenal venous blood, although being directly secreted by the adrenal cortex, does not respond to acute ACTH stimulation in contrast to the unconjugated steroid.

The highly significant correlations between basal levels of testosterone and oestrone may be explained by their common precursor and common sites of formation in the post-menopausal woman. Both arise mainly by peripheral conversion of 4-androstene-3,17-dione (to 60 and 100%, respectively) [32]. The formation of testosterone and oestrone occurs in the liver and adipose tissue, although catalysed by entirely different enzyme systems [33–35]. The significant correlations between DHA/DHAS and oestrone basal values reflect the exclusive adrenal origin of these steroids in the post-menopausal woman [15]. Significant correlations between basal levels of peripheral androgens and oestrogens have also been reported previously in post-menopausal women [16,32]. Although significant correlations between osteoporosis and low oestrogen levels have been reported by other workers [1–3,33] we found no significant correlation between bone mineral density and oestrone. However, we found a close association between DHA/DHAS and bone mineral density, which is in accordance with the results reported by Nordin and co-workers as to low levels of 4-androstene-3,17-dione and osteoporosis [2,3,32]. The association between bone mineral density and adrenal steroids seems to be restricted to the adrenal androgens since no correlations at all were found between cortisol and bone mineral density. The role of the age-related decrease in the levels of adrenal androgens ('adrenopause') in the aetiology of osteoporosis has previously been suggested by Nordin and co-workers [3,32]. Thus a decreased synthesis of adrenal androgens due to a 'premature adrenopause' in the post-menopausal women will lead to decreased oestrogen levels and finally to an increased rate of bone loss. It has been discussed whether or not the 'adrenopause' is a result of an isolated decline in the adrenal androgen production or the result of declining ACTH levels which maintain normal cortisol levels due to the age-related decrease in cortisol clearance [32]. The possibility of an age-related disappearance of the zona reticularis in the adrenal cortex, as has been suggested by Parker and Odell [36], as well as possible differences in the regulation of adrenal androgens and glucocorticoids as has been discussed above, may indicate the former as the more likely explanation.

The interesting finding of a significant correlation between bone mineral content and the condition of the uterine epithelium certainly reflects the differences in steroid hormone levels in women with high and low bone mineral density, respectively. Besides high bone mineral density and steroid levels in the upper range the post-menopausal women have been shown to be associated with endometrial abnormalities such as hyperplasia/atypia and cancer [6–11]. In a separate study on basal serum steroid levels in post-menopausal women with normal atrophic endometria and with benign hyperplasia/atypia, respectively, we have found significantly higher levels of DHA and DHAS in the latter group (Brody et al., unpubl. results).

The present study, although performed on limited material, has drawn further attention to a possible role of the adrenal cortex in the aetiology of osteoporosis and endometrial carcinoma, respectively. Further research is in progress to develop adreno-cortical function rests for identifying women belonging to those two risk groups.

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